Notes: Krueger (QJE, 1999)

Experimental Estimates of Education Production Functions

Contents

1 Big picture

- **Question.** What is the causal effect of reducing class size in early elementary grades on student achievement?
- **Setting.** The paper reanalyzes Tennessee Project STAR, a randomized experiment that assigned students and teachers to small classes, regular classes, or regular classes with a full-time aide from kindergarten through third grade.
- **Sample.** The STAR data include roughly 11,600 students across 80 schools, with 6,000–7,000 students observed in each year of the experiment.
- **Main outcome.** Achievement is measured primarily by average percentile scores on the Stanford Achievement Test, with robustness checks using Basic Skills First tests.
- **Main finding.** Students assigned to small classes score higher than students in regular-size classes. The first year in a small class produces a discrete gain of about four percentile points, and additional years add roughly one percentile point per year.
- **Teacher aides.** Regular classes with a full-time aide generally do not perform much better than regular classes without aides.
- **Experimental imperfections.** The paper addresses nonrandom attrition, re-randomization after kindergarten, switching between class types, variability in actual class size, and possible Hawthorne/John Henry effects.
- **Heterogeneity.** Small-class effects are larger for some minority, free-lunch, male, and school-location subsamples, although subgroup estimates are noisier.
- **Learning-process implication.** The major benefit appears to occur in the first year of small-class exposure, with smaller gains from continued exposure. This challenges pure value-added specifications that focus only on annual score gains.
- **Economic takeaway.** Well-designed experiments can discipline ambiguous observational evidence. In this case, smaller early-grade classes raise test performance, while teacher aides alone are not a close substitute for class-size reduction.

2 Data and measurement

2.1 Project STAR design

- Kindergarten students and teachers were randomly assigned within schools to three groups: small classes of about 13–17 students, regular classes of about 22–25 students, and regular classes with a full-time teacher aide.
- The design called for students to stay in the assigned class type from kindergarten through third grade.
- Every school was required to have at least one class of each type, so treatment comparisons are within-school comparisons.
- This is important because school quality, student demographics, and neighborhood conditions vary substantially across schools.

Interpretation. The within-school design gives the experiment much stronger identification than observational comparisons across schools or districts.

2.2 Outcomes

- The main dependent variable is the average percentile ranking on the Stanford Achievement Test.
- The average combines SAT math, reading, and word-recognition scores.
- Robustness checks use Basic Skills First outcomes and subject-specific test scores.
- Percentile-rank outcomes allow comparison across grades, but coefficients should be interpreted as percentile-point effects rather than raw-score effects.

Interpretation. A four percentile point increase means a shift in relative achievement rank, not a four percent increase in learning.

2.3 Treatment variables

- The reduced-form treatment variables are assignment to small class and assignment to regular/aide class, relative to regular class.
- Actual class size deviated from intended assignment because of mobility, administrative adjustments, and transitions across class types.
- The paper therefore distinguishes assignment effects from instrumental-variable estimates of actual class-size effects.
- For longitudinal models, the paper measures cumulative years spent in a small class.

Interpretation. Assignment is cleanly randomized; actual class size is the educational input. The paper uses both objects carefully.

2.4 Main threats to validity

- **Attrition:** some students leave the sample after initial assignment.
- **Re-randomization:** regular and regular/aide students were reassigned after kindergarten, while small-class students generally stayed in small classes.
- **Class switching:** about 10 percent of students switched between small and regular classes, often because of parental complaints or behavioral issues.
- **Actual class size variation:** realized class sizes were not perfectly fixed within intended ranges.
- **Behavioral responses:** treatment status might generate Hawthorne or John Henry effects.

Interpretation. The paper treats these deviations as econometric issues and tests whether the main results survive.

3 Models and empirical specifications

3.1 Reduced-form assignment model

$$Y_{ig} = \alpha_s + \gamma_g + \beta_1 Small_{ig} + \beta_2 Aide_{ig} + X_i'\delta + \varepsilon_{ig}. \tag{1}$$

Y_{ig} is a percentile score, α_s are school fixed effects, γ_g are grade or entry-grade controls, and X_i includes student controls where used.

Interpretation. The small-class coefficient estimates the intention-to-treat effect of being assigned to the small-class treatment.

3.2 Instrumental-variable model for actual class size

$$Y_{ig} = \alpha_s + \gamma_g + \theta ClassSize_{ig} + X_i'\delta + u_{ig}. \tag{2}$$

Actual class size is instrumented by randomized assignment to class type.

Interpretation. The IV model translates experimental assignment into the effect of a one-student change in actual class size around the STAR margin.

3.3 Pooled longitudinal model

$$Y_{ig} = \alpha_s + \gamma_g + \beta Small_{ig} + \rho YearsSmall_{ig} + X_i'\delta + \varepsilon_{ig}. \tag{3}$$

The small-class dummy captures the current level effect of being in a small class, while cumulative years in a small class captures the added effect of continued exposure.

Interpretation. A large β and smaller ρ imply a large initial gain followed by smaller cumulative gains, which is the paper's main timing pattern.

3.4 Heterogeneity and school-level effects

The paper estimates the pooled model separately by sex, free-lunch status, race, and school location. It also estimates school-specific small-class effects.

Interpretation. These exercises ask whether the average effect hides important differences across students and schools.

4 Identification and empirical design

4.1 Why the experiment is credible

- Randomization occurs within schools, so fixed school differences are removed by school fixed effects.
- Treatment arms are large and span many schools.
- Every school has all treatment arms, so comparisons are local within school.
- Parents and teachers did not choose initial class assignment.

Interpretation. The experiment improves on observational studies because treatment variation is generated by random assignment rather than family or school choice.

4.2 Why balance tests matter

Unadjusted groups differ on some observables because schools differ in composition. Within-school tests show that treatment assignment is not significantly related to most background variables.

Interpretation. The correct balance test is conditional on school because the experiment randomized within schools.

4.3 Attrition and noncompliance

Attrition and class switching could bias estimates if they are related to potential achievement. The paper shows that the small-class effect remains positive under alternative attrition and compliance adjustments.

Interpretation. The causal conclusion is strengthened because it survives deviations from the ideal experiment.

4.4 Limits of external validity

The experiment covers early elementary grades in Tennessee and identifies effects around a reduction of roughly seven to eight students per class. Effects may differ in later grades, other states, or poorly implemented programs.

Interpretation. The evidence is strong for the STAR margin, but not a universal class-size production function.

5 Tables and figures

5.1 Tables I and II: randomization balance

Read:

- Table I shows unadjusted means across small, regular, and regular/aide groups by entry wave.
- Unadjusted differences sometimes appear significant because randomization is within schools and schools differ in demographics.
- Table II reports within-school p-values for equality across class types.
- Within-school background variables are generally balanced, while actual class size is strongly different by design.

Interpretation. These tables justify the use of school fixed effects. Table I can look imbalanced because it ignores the level at which randomization occurred; Table II is the appropriate diagnostic for the experiment.

Economic message. The treatment-control comparison is credible once school effects are included.

What not to over-read. Do not treat unadjusted imbalance as evidence that the experiment failed; the relevant balance test is within schools.

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TABLE I
COMPARISON OF MEAN CHARACTERISTICS OF TREATMENTS AND CONTROLS: UNADJUSTED DATA

Variable	Small	Regular	Regular/Aide	Joint P-Value[a]
A. Students who entered STAR in kindergarten[b]				
1. Free lunch[c]	.47	.48	.50	.09
2. White/Asian	.68	.67	.66	.26
3. Age in 1985	5.44	5.43	5.42	.32
4. Attrition rate[d]	.49	.52	.53	.02
5. Class size in kindergarten	15.1	22.4	22.8	.00
6. Percentile score in kindergarten	54.7	49.9	50.0	.00
B. Students who entered STAR in first grade				
1. Free lunch	.59	.62	.61	.52
2. White/Asian	.62	.56	.64	.00
3. Age in 1985	5.78	5.86	5.88	.03
4. Attrition rate	.53	.51	.47	.07
5. Class size in first grade	15.9	22.7	23.5	.00
6. Percentile score in first grade	49.2	42.6	47.7	.00
C. Students who entered STAR in second grade				
1. Free lunch	.66	.63	.66	.60
2. White/Asian	.53	.54	.44	.00
3. Age in 1985	5.94	6.00	6.03	.66
4. Attrition rate	.37	.34	.35	.58
5. Class size in third grade	15.5	23.7	23.6	.01
6. Percentile score in second grade	46.4	45.3	41.7	.01
D. Students who entered STAR in third grade				
1. Free lunch	.60	.64	.69	.04
2. White/Asian	.66	.57	.55	.00
3. Age in 1985	5.95	5.92	5.99	.39
4. Attrition rate	NA	NA	NA	NA
5. Class size in third grade	16.0	24.1	24.4	.01
6. Percentile score in third grade	47.6	44.2	41.3	.01

a. p-value is for F-test of equality of all three groups.
b. Sample size in panel A ranges from 6299 to 6324, in panel B ranges from 2240 to 2314, in panel C ranges from 1585 to 1679, and in panel D ranges from 1202 to 1283.
c. Free lunch pertains to the fraction receiving a free lunch in the first year they are observed in the sample (i.e., in kindergarten for panel A; in first grade in panel B; etc.) Percentile score pertains to the average percentile score on the three Stanford Achievement Tests the students took in the first year they are observed in the sample.
d. Attrition rate is the fraction that ever exits the sample prior to completing third grade, even if they return to the sample in a subsequent year. Attrition rate is unavailable in third grade.

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TABLE II
P-VALUES FOR TESTS OF WITHIN-SCHOOL DIFFERENCES AMONG SMALL, REGULAR, AND REGULAR/AIDE CLASSES

Variable	Grade entered STAR program			
	K	1	2	3
1. Free lunch	.46	.29	.58	.18
2. White/Asian	.66	.28	.15	.21
3. Age	.38	.12	.48	.40
4. Attrition rate	.01	.07	.58	NA
5. Actual class size	.00	.00	.00	.00
6. Percentile score	.00	.00	.46	.00

5.2 Tables III and IV: actual class size and transitions

Read:

- Table III shows that actual class sizes largely follow intended assignment but overlap because of mobility and implementation variation.

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TABLE III
DISTRIBUTION OF CHILDREN ACROSS ACTUAL CLASS SIZES BY RANDOM ASSIGNMENT GROUP IN FIRST GRADE

Actual class size in first grade	Assignment group in first grade		
	Small	Regular	Aide
12	24	0	0
13	182	0	0
14	252	0	0
15	465	0	0
16	256	16	0
17	561	17	0
18	108	36	0
19	57	76	57
20	20	200	120
21	0	378	378
22	0	594	329
23	0	437	460
24	0	384	264
25	0	175	225
26	0	130	234
27	0	54	108
28	0	28	56
29	0	29	58
30	0	30	30
Average class size	15.7	22.7	23.4

Actual class was determined by counting the number of students in the data set with the same class identification.

It is also virtually impossible to prevent some students from switching between class types over time. Table IV shows a transition matrix between class types for students who continued from K–1, 1–2, and 2–3 grades. If students remained in their same class type over time, all the off-diagonal elements would be zero. The re-randomization of students in regular classes in first grade is apparent in panel A. But in second and third grades, when students were supposed to remain in their same type of class, 9–11 percent of students switched class-size types. Students were moved between class types because of behavioral problems or, in some cases, parental complaints. Obviously, if the movement between class types was associated with student characteristics

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TABLE IV
TRANSITIONS BETWEEN CLASS-SIZE IN ADJACENT GRADES
NUMBER OF STUDENTS IN EACH TYPE OF CLASS

A. Kindergarten to first grade

	First grade			
Kindergarten	Small	Regular	Reg/aide	All
Small	1292	60	48	1400
Regular	126	737	663	1526
Aide	122	761	706	1589
All	1540	1558	1417	4515

B. First grade to second grade

	Second grade			
First grade	Small	Regular	Reg/aide	All
Small	1435	23	24	1482
Regular	152	1498	202	1852
Aide	40	115	1560	1715
All	1627	1636	1786	5049

C. Second grade to third grade

	Third grade			
Second grade	Small	Regular	Reg/aide	All
Small	1564	37	35	1636
Regular	167	1485	152	1804
Aide	40	76	1857	1973
All	1771	1598	2044	5413

To address this potential problem, and the variability of class size for a given type of assignment, in some of the analysis that follows *initial* random assignment is used as an instrumental variable for actual class size.

B. Data and Standardized Tests

Students were tested at the end of March or beginning of April of each year. The tests consisted of the Stanford Achievement Test (SAT), which measured achievement in reading, word recognition, and math in grades K–3, and the Tennessee Basic Skills First (BSF) test, which measured achievement in reading and math in grades 1–3. The tests were tailored to each grade level. Because there are no natural units for the test results, I

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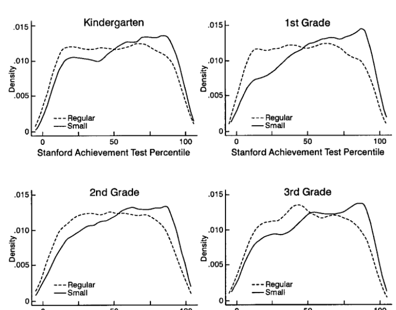

- Table IV shows transitions between class types across adjacent grades.
- Most students remain in their class type, but some switch between small, regular, and regular/aide classes.

Interpretation. These tables explain why the paper uses both reduced-form assignment estimates and IV estimates for actual class size. They also motivate cumulative exposure measures.

Economic message. The experiment created a large treatment contrast but had enough noncompliance to require careful econometrics.

What not to over-read. The overlap in actual class sizes does not invalidate the experiment; it changes how the treatment should be interpreted.

5.3 Figure I: score distributions

Read:

- The figure plots kernel densities of percentile scores by treatment group and grade.
- The small-class distribution is generally shifted to the right.
- The shift is visible across much of the distribution, not only at the mean.

Interpretation. The figure is a visual check that the treatment shifts the achievement distribution rather than simply changing a few outliers.

Economic message. Smaller classes improve the overall achievement distribution.

What not to over-read. The figure is descriptive; causal interpretation comes from random assignment and regression estimates.

5.4 Table V: reduced-form assignment effects

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whether the teacher has a master's degree—does not have a systematic effect. Hardly any of the teachers are male, so the gender results are not very meaningful. Teacher experience has a small, positive effect. Experimentation with a quadratic in experience indicated that the experience profile tends to peak at about

TABLE V
OLS AND REDUCED-FORM ESTIMATES OF EFFECT OF CLASS-SIZE ASSIGNMENT ON AVERAGE PERCENTILE OF STANFORD ACHIEVEMENT TEST

Explanatory variable	OLS: actual class size				Reduced form: initial class size			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. Kindergarten								
Small class	4.82	5.37	5.36	5.37	4.82	5.37	5.36	5.37
	(2.19)	(1.26)	(1.21)	(1.19)	(2.19)	(1.25)	(1.21)	(1.19)
Regular/aide class	.12	.29	.53	.31	.12	.29	.53	.31
	(2.23)	(1.13)	(1.09)	(1.07)	(2.23)	(1.13)	(1.09)	(1.07)
White/Asian (1 = yes	—	—	8.35	8.44	—	—	8.35	8.44
			(1.35)	(1.36)			(1.35)	(1.36)
Girl (1 = yes)	—	—	4.48	4.39	—	—	4.48	4.39
			(.63)	(.63)			(.63)	(.63)
Free lunch (1 = yes)	—	—	−13.15	−13.07	—	—	−13.15	−13.07
			(.77)	(.77)			(.77)	(.77)
White teacher	—	—	—	−.57	—	—	—	−.57
				(2.10)				(2.10)
Teacher experience	—	—	—	.26	—	—	—	.26
				(.10)				(.10)
Master's degree	—	—	—	−.51	—	—	—	−.51
				(1.06)				(1.06)
School fixed effects	No	Yes	Yes	Yes	No	Yes	Yes	Yes
R^2	.01	.25	.31	.31	.01	.25	.31	.31
B. First grade								
Small class	8.57	8.43	7.91	7.40	7.54	7.17	6.79	6.37
	(1.97)	(1.21)	(1.17)	(1.18)	(1.76)	(1.14)	(1.10)	(1.11)
Regular/aide class	3.44	2.22	2.23	1.78	1.92	1.69	1.64	1.48
	(2.05)	(1.00)	(0.98)	(0.98)	(1.12)	(0.80)	(0.76)	(0.76)
White/Asian (1 = yes)	—	—	6.97	6.97	—	—	6.86	6.85
			(1.18)	(1.19)			(1.18)	(1.18)
Girl (1 = yes)	—	—	3.80	3.85	—	—	3.76	3.82
			(.56)	(.56)			(.56)	(.56)
Free lunch (1 = yes)	—	—	−13.49	−13.61	—	—	−13.65	−13.77
			(.87)	(.87)			(.88)	(.87)
White teacher	—	—	—	−4.28	—	—	—	−4.40
				(1.96)				(1.97)
Male teacher	—	—	—	11.82	—	—	—	13.06
				(3.33)				(3.38)
Teacher experience	—	—	—	.05	—	—	—	.06
				(0.06)				(.06)
Master's degree	—	—	—	.48	—	—	—	.63
				(1.07)				(1.09)
School fixed effects	No	Yes	Yes	Yes	No	Yes	Yes	Yes
R^2	.02	.24	.30	.30	.01	.23	.29	.30

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TABLE V
(CONTINUED)

Explanatory variable	OLS: actual class size				Reduced form: initial class size			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
C. Second grade								
Small class	5.93	6.33	5.83	5.79	5.31	5.52	5.27	5.26
	(1.97)	(1.29)	(1.23)	(1.23)	(1.70)	(1.16)	(1.10)	(1.10)
Regular/aide class	1.97	1.88	1.64	1.58	.47	1.44	1.16	1.18
	(2.05)	(1.10)	(1.07)	(1.06)	(1.23)	(0.87)	(0.81)	(0.81)
White/Asian (1 = yes)	—	—	6.35	6.36	—	—	6.27	6.29
			(1.20)	(1.19)			(1.21)	(1.20)
Girl (1 = yes)	—	—	3.48	3.45	—	—	3.48	3.44
			(.60)	(.60)			(.60)	(.60)
Free lunch (1 = yes)	—	—	−13.61	−13.61	—	—	−13.75	−13.77
			(.72)	(.72)			(.73)	(.73)
White teacher	—	—	—	.39	—	—	—	.43
				(1.75)				(1.76)
Male teacher	—	—	—	1.32	—	—	—	.82
				(3.96)				(4.23)
Teacher experience	—	—	—	.10	—	—	—	.10
				(.06)				(.07)
Master's degree	—	—	—	−1.06	—	—	—	−1.16
				(1.06)				(1.05)
School fixed effects	No	Yes	Yes	Yes	No	Yes	Yes	Yes
R^2	.01	.22	.28	.28	.01	.21	.28	.28
D. Third grade								
Small class	5.32	5.58	5.01	5.00	5.51	5.42	5.30	5.24
	(1.91)	(1.22)	(1.19)	(1.19)	(1.46)	(1.08)	(1.03)	(1.04)
Regular/aide class	−.22	−.16	−.33	−.75	−.30	.12	.13	−.10
	(1.95)	(1.12)	(1.11)	(1.07)	(1.17)	(0.85)	(0.81)	(0.78)
White/Asian (1 = yes)	—	—	6.12	6.11	—	—	5.97	5.96
			(1.45)	(1.44)			(1.44)	(1.43)
Girl (1 = yes)	—	—	4.16	4.16	—	—	4.17	4.18
			(.66)	(.65)			(.66)	(.66)
Free lunch (1 = yes)	—	—	−13.02	−12.96	—	—	−13.21	−13.16
			(.81)	(.81)			(.82)	(.81)
White teacher	—	—	—	.64	—	—	—	.19
				(1.75)				(1.75)
Male teacher	—	—	—	−7.42	—	—	—	−6.83
				(2.80)				(2.76)
Teacher experience	—	—	—	.04	—	—	—	.03
				(.06)				(.06)
Master's degree	—	—	—	1.10	—	—	—	.88
				(1.15)				(1.15)
School fixed effects	No	Yes	Yes	Yes	No	Yes	Yes	Yes
R^2	.01	.17	.22	.23	.01	.16	.22	.22

All models include constants. Robust standard errors that allow for correlated residuals among students

Read:

- Small-class assignment has positive and statistically significant effects across specifications.
- Regular/aide effects are generally modest relative to small-class effects.
- The result survives adding controls, school effects, and alternative sample restrictions.

Interpretation. Table V is the core experimental estimate. It shows that the small-class result is not produced by one fragile specification.

Economic message. Reducing class size changes achievement more than adding a teacher aide to a regular class.

What not to over-read. This is an intention-to-treat effect, not exactly the effect of one fewer student in a classroom.

5.5 Tables VI–VIII: attrition and actual class-size effects

Read:

- Table VI checks attrition and sample restrictions.
- Table VII estimates the effect of actual class size using randomized assignment as an instrument.
- Table VIII shows how actual class-size effects vary by entry grade and current grade.

Interpretation. These tables show that the main result is robust to missing data and imperfect implementation. They also translate assignment into actual class-size effects.

Economic message. Smaller actual classes matter, not only assignment labels.

What not to over-read. The IV effect is local to class sizes generated by STAR and should not be extrapolated mechanically to all class-size changes.

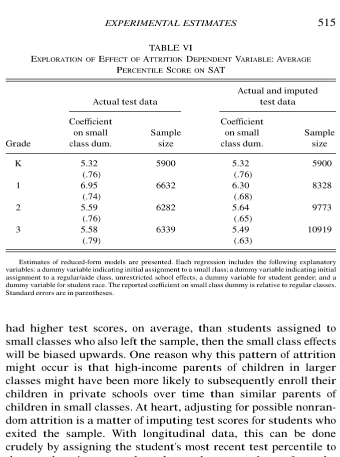

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TABLE VI
EXPLORATION OF EFFECT OF ATTRITION DEPENDENT VARIABLE: AVERAGE PERCENTILE SCORE ON SAT

Grade	Actual test data: Coefficient on small class dum.	Actual test data: Sample size	Actual and imputed test data: Coefficient on small class dum.	Actual and imputed test data: Sample size
K	5.32	5900	5.32	5900
	(.76)		(.76)	
1	6.95	6632	6.30	8328
	(.74)		(.68)	
2	5.59	6282	5.64	9773
	(.76)		(.65)	
3	5.58	6339	5.49	10919
	(.79)		(.63)	

Estimates of reduced-form models are presented. Each regression includes the following explanatory variables: a dummy variable indicating initial assignment to a small class; a dummy variable indicating initial assignment to a regular/aide class, unrestricted school effects; a dummy variable for student gender; and a dummy variable for student race. The reported coefficient on small class dummy is relative to regular classes. Standard errors are in parentheses.

had higher test scores, on average, than students assigned to small classes who also left the sample, then the small class effects will be biased upwards. One reason why this pattern of attrition might occur is that high-income parents of children in larger classes might have been more likely to subsequently enroll their children in private schools over time than similar parents of children in small classes. At heart, adjusting for possible nonrandom attrition is a matter of imputing test scores for students who exited the sample. With longitudinal data, this can be done crudely by assigning the student's most recent test percentile to that student in years when the student was absent from the

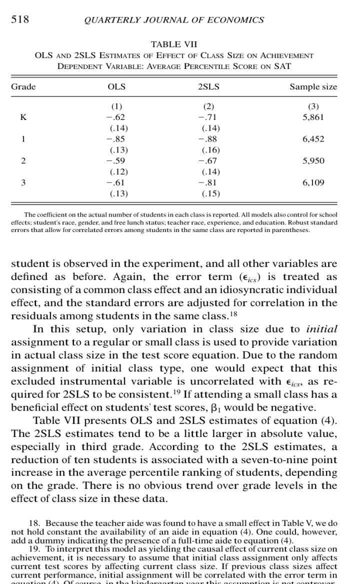

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TABLE VII
OLS AND 2SLS ESTIMATES OF EFFECT OF CLASS SIZE ON ACHIEVEMENT
DEPENDENT VARIABLE: AVERAGE PERCENTILE SCORE ON SAT

Grade	OLS	2SLS	Sample size
	(1)	(2)	(3)
K	−.62	−.71	5,861
	(.14)	(.14)	
1	−.85	−.88	6,452
	(.13)	(.16)	
2	−.59	−.67	5,950
	(.12)	(.14)	
3	−.61	−.81	6,109
	(.13)	(.15)	

The coefficient on the actual number of students in each class is reported. All models also control for school effects; student's race, gender, and free lunch status; teacher race, experience, and education. Robust standard errors that allow for correlated errors among students in the same class are reported in parentheses.

student is observed in the experiment, and all other variables are defined as before. Again, the error term (ϵ_{ics}) is treated as consisting of a common class effect and an idiosyncratic individual effect, and the standard errors are adjusted for correlation in the residuals among students in the same class.[18]

In this setup, only variation in class size due to *initial* assignment to a regular or small class is used to provide variation in actual class size in the test score equation. Due to the random assignment of initial class type, one would expect that this excluded instrumental variable is uncorrelated with ϵ_{ics}, as required for 2SLS to be consistent.[19] If attending a small class has a beneficial effect on students' test scores, β_1 would be negative.

Table VII presents OLS and 2SLS estimates of equation (4). The 2SLS estimates tend to be a little larger in absolute value, especially in third grade. According to the 2SLS estimates, a reduction of ten students is associated with a seven-to-nine point increase in the average percentile ranking of students, depending on the grade. There is no obvious trend over grade levels in the effect of class size in these data.

18. Because the teacher aide was found to have a small effect in Table V, we do not hold constant the availability of an aide in equation (4). One could, however, add a dummy indicating the presence of a full-time aide to equation (4).

19. To interpret this model as yielding the causal effect of current class size on achievement, it is necessary to assume that initial class assignment only affects current test scores by affecting current class size. If previous class sizes affect current performance, initial assignment will be correlated with the error term in equation (4). Of course, in the kindergarten year this assumption is not controver-

5.6 Table IX: pooled longitudinal models

Read:

- Table IX separates the initial small-class effect from cumulative years in small classes.
- The initial effect is large; the cumulative-years effect is positive but smaller.

Interpretation. The timing pattern suggests a one-time level effect from initial exposure, with smaller additional gains from continued exposure.

Economic message. Value-added models focused only on annual gains may miss the main benefit of small classes.

What not to over-read. The table does not prove the exact mechanism behind the initial jump.

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Table VIII presents additional 2SLS estimates of the effect of actual class size on achievement, disaggregating the sample by the grade the student entered Project STAR and current grade. The model and identification strategy are the same as in Table VII, column 2. The results indicate that for each cohort of students, those attending smaller classes tend to score higher on the standardized test by the end of the *first* year they entered the experiment. If assignment to small or regular classes was somehow nonrandom, then the initial assignment would have to have been skewed in the direction of producing higher test scores in the small classes for each wave of students who entered the program—an unlikely event. Interestingly, for the wave of students who entered in kindergarten, the beneficial effect of attending a small class does not appear to increase as students spend more time in their class assignment. For students entering the experiment in first or second grade, however, the test score gap between those in small- and regular-size classes grows as students progress to higher grades. The effect of time spent in a small class is explored further by pooling students in all grades together below.

C. Models with Pooled Data

To explore the cumulative effects of having been in a small or regular class, several models were estimated with the data pooled

TABLE VIII
2SLS ESTIMATES OF EFFECT OF CLASS SIZE ON ACHIEVEMENT, BY ENTRY GRADE AND CURRENT GRADE
DEPENDENT VARIABLE: AVERAGE SCORE ON STANFORD ACHIEVEMENT TEST

Current grade	Entering grade			
	K	1	2	3
K	−.71			
	(.15)			
1	−.89	−.49		
	(.17)	(.23)		
2	−.49	−.70	−.24	
	(.16)	(.29)	(.21)	
3	−.66	−1.21	−.71	−.66
	(.17)	(.34)	(.28)	(.21)

The coefficient on the actual number of students in each class is reported. All models also control for school effects; student's race, gender, and free lunch status; teacher race, experience, and education. Robust standard errors that allow for correlation of residuals among students in the same class are reported in parentheses. Sample size in column 1 begins at 5901 and ends at 3124; sample size in column 2 begins at 2190 and ends at 1110; sample size in column 3 begins at 1492 and ends at 1010; sample size in column 4 is 1110.

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TABLE IX
ESTIMATES OF POOLED MODELS
DEPENDENT VARIABLE: AVERAGE PERCENTILE RANKING ON SAT TEST
COEFFICIENT ESTIMATES WITH ROBUST STANDARD ERRORS IN PARENTHESES

Variable	(1)	(2)	(3)
Initial class small (1 = yes)	2.87	3.16	2.99
	(.83)	(.80)	(.80)
Initial class regular/aide (1 = yes)	.29	.49	.58
	(.69)	(.67)	(.67)
Cumulative years in small class	1.19	1.05	.65
	(.39)	(.38)	(.39)
Cumulative years in reg/aide class	.37	.25	.14
	(.39)	(.37)	(.37)
Fraction of classmates in class previous year	—	—	.60
			(1.03)
Average fraction of classmates together previous year	—	—	−.46
			(1.52)
Fraction of classmates on free lunch	—	—	−2.73
			(1.62)
Fraction of classmates who attended kindergarten	—	—	6.85
			(1.67)
Student and teacher characteristics	No	Yes	Yes
3 current grade dummies; 3 dummies indicating first grade appeared in sample; school effects	Yes	Yes	Yes
R^2	.18	.23	.23
Sample size	25,249	24,350	24,349

Student and teacher characteristics are as follows: student race, gender, and free lunch status; and teacher race, gender, experience, and master's degree or higher status. OLS estimates are reported, with robust standard errors that adjust for a possible correlation of residuals for the same student over time in parentheses.

In addition to these two "class constancy" variables, the regression includes the fraction of students in a class who receive free lunch and the fraction of students in the class who were present in the experiment during kindergarten. Because students on free lunch score lower on standardized tests than other students, a higher proportion of classmates on free lunch in a class may lower overall performance. The fraction of a class that attended kindergarten could affect achievement because kindergarten attendance is likely to make the class more socialized for school, which should enable the teacher to convey more material. Due to the random assignment of students, these variables should be uncorrelated with any omitted variables within schools.

Including these four variables hardly changes the initial jump in test scores associated with attending a small class (see column 3), although the cumulative effect of time spent in a small

5.7 Table X and Figure II: heterogeneity

Read:

- Table X estimates the model separately by sex, lunch status, race, and school location.
- Small-class effects are positive in most subgroups and appear larger for some disadvantaged groups.
- Figure II shows that school-level effects vary substantially, with about two-thirds positive.

Interpretation. The average effect hides meaningful heterogeneity. Some schools translate smaller classes into gains more effectively than others.

Economic message. Class-size policy may be most valuable where teacher attention is most scarce or where implementation is strong.

What not to over-read. Subgroup estimates are less precise and involve many comparisons, so treat differences as suggestive.

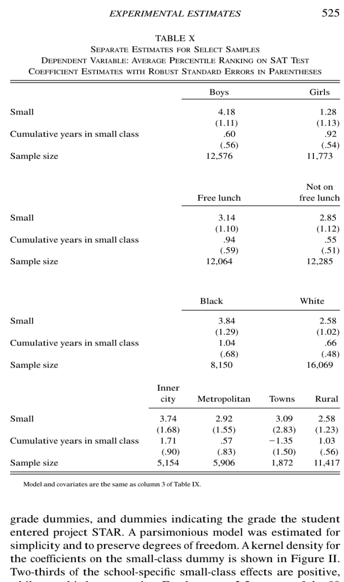

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TABLE X
SEPARATE ESTIMATES FOR SELECT SAMPLES
DEPENDENT VARIABLE: AVERAGE PERCENTILE RANKING ON SAT TEST
COEFFICIENT ESTIMATES WITH ROBUST STANDARD ERRORS IN PARENTHESES

	Boys	Girls
Small	4.18	1.28
	(1.11)	(1.13)
Cumulative years in small class	.60	.92
	(.56)	(.54)
Sample size	12,576	11,773

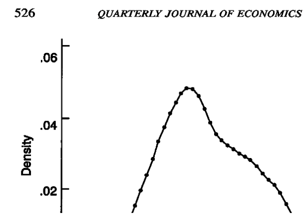
	Free lunch	Not on free lunch
Small	3.14	2.85
	(1.10)	(1.12)
Cumulative years in small class	.94	.55
	(.59)	(.51)
Sample size	12,064	12,285

	Black	White
Small	3.84	2.58
	(1.29)	(1.02)
Cumulative years in small class	1.04	.66
	(.68)	(.48)
Sample size	8,150	16,069

	Inner city	Metropolitan	Towns	Rural
Small	3.74	2.92	3.09	2.58
	(1.68)	(1.55)	(2.83)	(1.23)
Cumulative years in small class	1.71	.57	−1.35	1.03
	(.90)	(.83)	(1.50)	(.56)
Sample size	5,154	5,906	1,872	11,417

Model and covariates are the same as column 3 of Table IX.

grade dummies, and dummies indicating the grade the student entered project STAR. A parsimonious model was estimated for simplicity and to preserve degrees of freedom. A kernel density for the coefficients on the small-class dummy is shown in Figure II. Two-thirds of the school-specific small-class effects are positive,



5.8 Table XI and appendix statistics

Read:

- Table XI shows positive small-class effects across SAT math, reading, word recognition, and BSF outcomes.
- The appendix summary statistics describe class size, percentile scores, free-lunch share, race, gender, age, attrition, teacher credentials, schools, students, and classes by grade.

Interpretation. The subject-test results show that the average SAT outcome is not hiding a narrow effect in just one subject. The appendix table situates the sample and verifies the scale of STAR.

Economic message. Small classes have broad achievement effects across tests.

What not to over-read. These are still test-score outcomes; long-run economic effects require additional assumptions.

6 Short summary

- Project STAR is a randomized experiment on early-grade class size.
- Within-school balance tests support the validity of random assignment.
- Small classes raise standardized test performance by several percentile points.
- Teacher aides in regular-size classes have only modest effects.
- Attrition, switching, re-randomization, and actual class-size variation do not overturn the result.
- The first year in a small class appears especially important.
- Effects vary across students and schools, suggesting implementation and targeting matter.
- The rough cost-benefit calculation suggests benefits may be in the same ballpark as costs.

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TABLE XI
ESTIMATES OF POOLED DATA MODEL BY SUBJECT TEST
DEPENDENT VARIABLE: PERCENTILE SCORE ON SAT OR BSF TEST
COEFFICIENT ESTIMATES WITH ROBUST STANDARD ERRORS IN PARENTHESES

	Stanford Achievement Test			Basic Skills First		
	Math	Reading	Word	Math	Reading	Avg.
Small	2.83	3.52	2.97	1.09	3.04	2.02
	(.88)	(.88)	(.87)	(1.05)	(1.08)	(.96)
Cumulative years in small class	.45	.43	.80	1.23	.41	.83
	(.42)	(.43)	(.42)	(.44)	(.46)	(.41)
Sample size	23,794	23,461	23,630	18,174	18,010	18,250

Model and covariates are the same as in column 3 of Table IX.

reducing class size may be different than assumed. There is no substitute for directly measuring the economic outcomes that may be affected by reducing class size. Nonetheless, these calculations suggest that the benefit of reducing class size in terms of future earnings is in the same ballpark as the costs.

APPENDIX: SUMMARY STATISTICS
MEANS WITH STANDARD DEVIATIONS IN PARENTHESES

	Grade				
Variable	K	1	2	3	All
Class size	20.3	21.0	21.1	21.3	20.9
	(4.0)	(4.0)	(4.1)	(4.4)	(4.1)
Percentile score avg. SAT	51.4	51.5	51.2	51.0	51.3
	(26.7)	(26.9)	(26.5)	(27.0)	(26.8)
Percentile score avg. BSF	NA	51.8	51.6	51.4	51.6
		(26.1)	(26.2)	(26.1)	(26.1)
Free lunch	.48	.52	.51	.50	.51
White	.67	.67	.65	.66	.66
Girl	.49	.48	.48	.48	.47
Age[a]	5.43	6.58	7.67	8.70	7.12
	(0.35)	(0.49)	(0.56)	(0.59)	(1.31)
Exited sample[b]	.29	.26	.21	NA	.43
Retained	NA	NA	NA	.04	NA
Percent of teachers[c] with MA+ degree	.35	.35	.37	.44	.38
Percent of teachers who are White	.83	.82	.79	.79	.81
Percent of teachers who are male	.00	.00	.01	.03	.01
No. of schools	79	76	75	75	80
No. of students	6,323	6,828	6,839	6,801	11,599
No. of small classes	127	124	133	140	524
No. of reg. classes	99	115	100	89	403
No. of reg/aide classes	99	100	107	107	413

a. Age as of September of the school year they are observed.
b. The fraction that exited the sample in the next year, for K–2; for All it is the fraction that ever exited the sample.
c. Teacher characteristics are weighted by the number of students in each teacher's class.

7 Expanded conclusion

7.1 Core conclusion

Krueger's reanalysis of Project STAR concludes that reducing class size in early grades raises student achievement. The finding survives adjustments for school effects, attrition, re-randomization, nonrandom transitions, and actual class-size variation.

7.2 What is learned about education production functions

The paper shows that observational education production functions can be fragile because class size, student composition, teacher quality, and school quality are jointly determined. Random assignment makes Project STAR unusually informative.

7.3 Why timing matters

The main benefit appears to occur in the first year of small-class exposure, with additional years adding smaller gains. This challenges value-added specifications that focus only on year-to-year gains.

7.4 Policy interpretation

The evidence is strongest for early elementary grades and for reductions near the STAR margin. It supports class-size reduction as an educational input but does not imply that every class-size policy everywhere will be cost effective.

7.5 Teacher aides versus smaller classes

The aide results imply that adding another adult to a large classroom is not an equivalent substitute for reducing the number of students per teacher.

7.6 Cost-benefit implications

Krueger's rough calculation suggests that if the test-score gains persist and translate into earnings gains, the benefits of smaller classes may be comparable to the costs. This is not a final welfare calculation, but it shows the effects are economically meaningful.

7.7 Final takeaway

The enduring lesson is both substantive and methodological: smaller classes in early grades can raise achievement, and randomized experiments can clarify debates that observational evidence leaves unresolved.